

Emelio Joshia Exaudi Lumban Gaol

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Final-year Meteorology student at Institut Teknologi Bandung with hands-on experience in data analysis and statistical modeling. Proficient in **Python, SQL, and Microsoft Excel**, and comfortable working with large, noisy environmental datasets across mining and climate sectors. Dean's List scholar (Semester 7 GPA 4.00 / 4.00) with experience leading a 10-person technical team. Actively seeking opportunities to apply analytical skills and a scientific mindset across industries.

EDUCATION

Institut Teknologi Bandung	<i>Bandung, Indonesia</i>
Bachelor of Science in Meteorology	<i>Aug 2022 – Present</i>
• GPA: 3.14 / 4.00 Dean's List (Semester 7): 4.00 / 4.00. • Undergraduate Thesis: Analyzing land-use change impact on river discharge in DAS Sepaku (IKN region) using WRF-Hydro simulations on an HPC cluster, processing 1-km-resolution geospatial grids and multi-gigabyte atmospheric forcing data. • Relevant Coursework: Data Science, Statistical Modeling, Computational Methods, Atmospheric Modeling, Hydrometeorology, Weather Prediction.	

EXPERIENCE

PT Aneka Tambang Tbk (ANTAM) UBPE Pongkor	<i>Bogor, Indonesia</i>
Health, Safety, Security & Environment Intern	<i>Jan 2026 – Feb 2026</i>
• Performed bias correction on two satellite rainfall products (GPM IMERG, CHIRPS) against ground rain-gauge data using Linear Scaling and Quantile Mapping in Python, benchmarking method performance to identify the best-performing combination for site-level analysis. • Analyzed rainfall-to-movement lag time at two extensometer stations using cross-correlation analysis, surfacing significant patterns that informed slope-monitoring recommendations. • Evaluated the operational siren threshold against one year of field data and recommended antecedent-rainfall and multi-day cumulative approaches as improvements to the existing early-warning system. • Conducted spatial analysis of the mine concession area in QGIS using DEMNAS to produce elevation and slope-classification maps that contextualize landslide susceptibility. • Presented four operational recommendations to HSSE department leadership, covering data-collection protocols, weather station network expansion, location-specific forecast tools, and a monitoring dashboard architecture.	
Indonesian Agency for Meteorology, Climatology, and Geophysics (BMKG)	<i>Jakarta, Indonesia</i>
Climate Change Analysis Intern	<i>Jul 2025 – Aug 2025</i>
• Performed statistical downscaling of CMIP6 global climate model outputs using Linear Scaling bias correction in Python, generating high-resolution projection datasets for climate-risk assessment in the mining sector. • Validated downscaled outputs against CHIRPS satellite rainfall records to quantify residual bias and assess method performance. • Built Python workflows using xarray and netCDF4 to process multi-gigabyte climate datasets, and presented methodology and findings to the Climate Change Information Centre team.	
Institut Teknologi Bandung Applied Meteorology Laboratory	<i>Bandung, Indonesia</i>
Laboratory Assistant	<i>Apr 2024 – Present</i>
• Assist in running weekly practicum sessions for undergraduate students, including data-collection exercises, instrument operation tutorials, and result analysis. • Maintain and calibrate meteorological observation instruments (rain gauges, thermometers, anemometers, barometers, weather stations) to ensure data reliability for academic and research use.	
Institut Teknologi Bandung	<i>Bandung, Indonesia</i>
Teaching Assistant Continuum Mechanics & Observational Meteorology	<i>Sep 2024 – Sep 2025</i>
• Led weekly tutorial and discussion sessions for two undergraduate courses, covering problem-solving in fluid dynamics, vector calculus, and observational data interpretation. • Designed practicum modules and assessment rubrics, graded assignments, and provided technical office hours to support students in mastering core concepts.	

SELECTED PROJECTS

ECOS: EV Battery Circular Ownership System | IEEEBIG 2026 Business Case Competition (Finalist)

- Co-developed an **end-of-life EV battery management framework** (Battery-as-a-Service deposit scheme, SPKLU collection network, battery grading facility) for an Indonesian battery manufacturer; quantified **12 operational and financial risk factors** through a **risk-assessment heatmap (Python, matplotlib)** and contributed to financial-viability analysis that anchored the team's submission to the national finals.

LEADERSHIP & ACTIVITIES

Atmosphaira, Meteorology Student Association ITB

Bandung, Indonesia

Head of Research and Development Division

May 2025 – Feb 2026

- Led a **10-person cross-functional team** through the full product-development cycle of an **IoT-based Automatic Water Level Recorder (AWLR)** and **Automatic Rainfall Recorder (ARR)**, from hardware prototyping and firmware development through **field deployment as early-warning tools** in a partner village.
- Built **hydrometeorological models using WRF-Hydro** to simulate rainfall accumulation and river discharge, producing predictive maps that informed local disaster preparedness planning.
- Managed task delegation, **weekly sprint reviews**, and stakeholder coordination with faculty supervisors and external partners; conducted a formal **post-mortem** to surface lessons for the incoming division.

Intracampus Division Staff

May 2024 – Apr 2025

- Organized **cross-association study visits** and collaborative student programs to strengthen inter-faculty networks within the FITB school.

SKILLS & TOOLS

Hard Skills: Python (pandas, NumPy, scikit-learn, matplotlib, xarray), SQL, Microsoft Excel, MATLAB, Bash; statistical analysis, bias correction, time-series and lag analysis, regression, classification, exploratory data analysis (EDA).

Tools & Modeling: ArcGIS, QGIS, Google Earth Engine; WRF-Hydro, HEC-RAS, HEC-HMS, CMIP6 downscaling.

Soft Skills: Problem-solving, communication, teamwork, leadership, adaptability, willingness to learn, critical thinking, time management.

Languages: Indonesian (native), English (professional working proficiency).

Certifications: Google Data Analytics Professional Certificate (Coursera); Google Cybersecurity Professional Certificate (Coursera); Linux, SQL, and Python automation modules; CARE for SEA and CORDEX-SEA training (World Climate Research Programme).